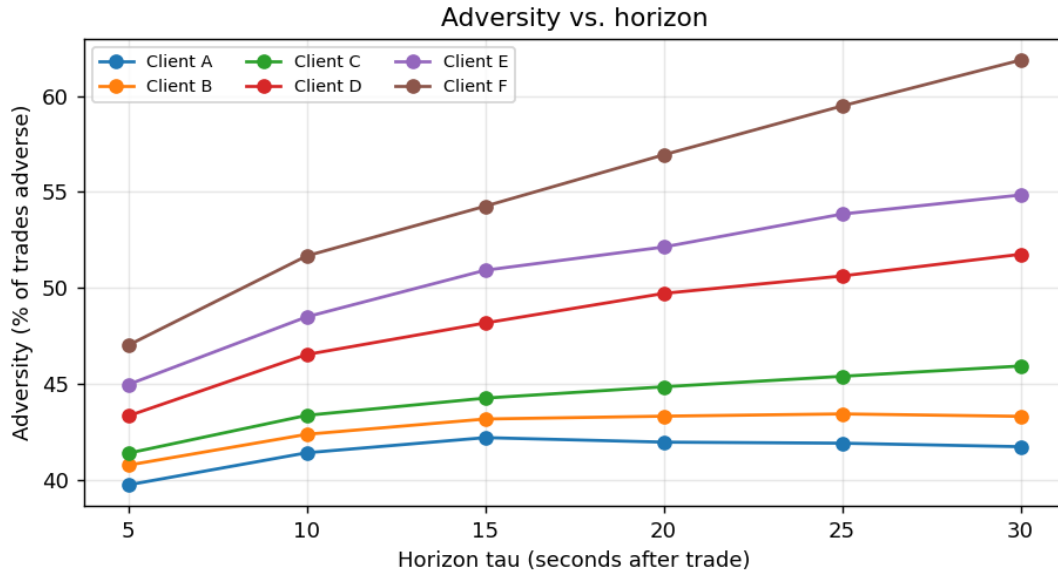


Task 1 - Adversity Profiles of Each Client

A trade is adverse at horizon τ if, closing the position at $t = \tau$, the LP PnL is negative: $\text{side} \cdot V \cdot (M_\tau - T_P) < 0$. Since $V > 0$, only the sign of $\text{side} \cdot (M_\tau - T_P)$ matters. Adversity(client, τ) = % of that client's trades that are adverse.



Results table (adversity %, by horizon)

client	$\tau=5$	$\tau=10$	$\tau=15$	$\tau=20$	$\tau=25$	$\tau=30$
A	39.72	41.40	42.19	41.95	41.90	41.71
B	40.75	42.36	43.16	43.30	43.42	43.30
C	41.37	43.35	44.25	44.84	45.39	45.92
D	43.32	46.52	48.17	49.71	50.62	51.75
E	44.96	48.49	50.92	52.13	53.85	54.84
F	47.01	51.67	54.26	56.94	59.49	61.88

Trends observed

- 1. Monotone in horizon. For every client adversity rises (or is flat) as τ grows from 5s to 30s. Right after a fill the mid is still close to T_P , so roughly half the trades look adverse; as time passes informed flow keeps pushing the mid against us, so more positions would close at a loss. This is the signature of adverse selection / short-horizon drift.
- 2. Clear ranking across clients. A and B stay near 40-43% and almost flat - close to a coin flip, i.e. little information in their flow. C is mildly adverse. D, E and F are progressively worse, with F climbing from ~47% at 5s to ~62% at 30s. The steeper the curve, the more informed (toxic) the client: their trades systematically precede mid moves against us.
- 3. Interpretation for the LP. Clients whose adversity exceeds 50% and grows with τ (E, F, and D at long horizons) are picking us off; clients below 50% (A, B, C) are benign. This ranking directly motivates the per-client spread widening quantified in Task 2.